

Udbhav Balaji

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PROFESSIONAL SUMMARY

As a recent graduate from Queen's University with over 6 years of experience in Python and SQL, I have honed my skills in Javascript, React and Typescript, with a keen interest in full stack development. My passion for data science and machine learning is exemplified through a supervised learning project based on Spotify's song data, showcasing my commitment to problem-solving and making a real-world impact.

EDUCATION

Queen's University

Bachelor of Computing (Honours.) in Computer Science

Kingston, ON

- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Artificial Intelligence
- **Final GPA & Awards:** 3.31; Dean's Honour List

WORK EXPERIENCE

Celestica

05/2022 - 08/2023

Data Analyst & Quote Controller Intern

Toronto, ON

- Facilitated spend data classification to correctly identify categories of spend, reducing unclassified spend by 30%
- Improved ease of compliance by identifying critical contract terms and metrics to assist the transition between contract management systems, enabling greater visibility of contract terms
- Built a PowerBI dashboard to enable senior management in evolving appropriate policies for effective management of employee travel expenses

Queen's University

01/2022 - 04/2022 & 01/2024 - 04/2024

Teaching Assistant

Kingston, ON

- **Courses:** Linear Data Analysis, Advanced Data Analytics
- **Responsibilities:** Grading, Office Hours, Tutorial Session Mentoring

Queen's University

08/2021 - 04/2022

Residence Don (Resident Assistant)

Kingston, ON

- Promoted academic success, addressed student needs, encouraged involvement among students in residence
- Implemented and enforced university and campus policies
- Conducted campus events to boost community engagement while maintaining student safety during the pandemic

PROJECT WORK

[Catch-Result](#) | *Typescript, bun, [npm package](#)*

- Engineered a type-safe error-handling library in **Typescript** that simplifies error management, greatly improving code clarity with a fluent API
- Built-in granular error matching to allow catching of specific error types
- Published to npm with full type definitions, documentation and test cases, demonstrating strong API design and library development skills

[Quiklist](#) | *Typescript, Node, [npm package](#)*

- Built a high-performance, terminal-native checklist application for efficient command-line task management
- Developed intuitive CLI commands for adding, viewing, editing, and deleting tasks.
- Designed for full customizability with local JSON data persistence and optional Git-based synchronization.
- Emphasized fast UX and minimal dependencies to support developer productivity within terminal workflows.

[Spotify Hit Predictor](#) | *Python, Jupyter, Neural Network, Machine Learning, MLPClassifier*

- Developed a classification model to predict song success based on Spotify's track metrics
- Achieved an overall accuracy of ~88% with multi layer perceptron after cross validation
- Has been deprecated due to [change in Spotify's API access](#) to retrieve track features

[Rubiks Cube Architecture](#) | *Python, Object Oriented Programming (OOP), numpy*

- Built Rubix Cube structure/architecture provisioned to perform all operations defined by a real-world cube using Inheritance and Object-Oriented Programming (OOP) as precursor to a Rubik's Cube Solver
- Built-in command line interface to interact with cube instance (perform operations & visualize cube)

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, Typescript, PHP, SQL, HTML/CSS, MATLAB

Frameworks: Express, React, Flask, Tailwind, Next.js, React Native, Expo

Libraries: pandas, numpy, matplotlib, scikit-learn, scipy, SQLAlchemy, Flask

Technologies: Node.js, REST APIs, MySQL, Postgres, Git, Vercel, MS Excel, PowerBI, Bun